Data Structures & Algorithms

BRANCH: COMPUTER SCIENCE (CSE)
COURSE CODE: CS201

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Data Structures & Algorithms (CS201) under the Computer Science (CSE) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Linear Data Structures

Introduction to complexity analysis. Big-O notation. Arrays, Linked Lists (Singly, Doubly, Circular), Stacks, and Queues. Applications of stacks: Infix to Postfix conversion, recursion evaluation.

Unit 2: Non-Linear Data Structures

Trees: Binary Trees, Binary Search Trees (BST), AVL Trees, Red-Black Trees, Heaps, and B-Trees. Graphs: Representation (Adjacency Matrix, Adjacency List), Traversals (BFS, DFS).

Unit 3: Sorting & Searching

Searching: Linear Search, Binary Search, Hashing. Sorting: Bubble Sort, Selection Sort, Insertion Sort, Quick Sort, Merge Sort, Heap Sort. Time and space complexity comparisons.

Unit 4: Advanced Algorithms

Greedy Algorithms (Huffman coding, Prim's and Kruskal's MST). Dynamic Programming (Knapsack, LCS). Backtracking (N-Queens, Graph Coloring).

Key Formulas & Exam Pointers

CRITICAL FORMULA / PRINCIPLE:

Big-O time complexity, BST search time $O(\log n)$, AVL height-balance factor $|h_L - h_R| <= 1$.